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Systematic experiments for proof of Poisson statistics on direct-detection laser radar using Geiger mode avalanche photodiode

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ABSTRACT

The single pulse detection and false-alarm probability in measuring time of flight (TOF) of laser pulse are caused by the creation of primary electrons in Geiger mode avalanche photodiode (APD). Measuring the detection and false-alarm probabilities with longer laser pulse than time bin and artificial noise source, we have experimentally proved that the creations of primary electrons in Geiger mode APD is Poisson distributed. It is also shown theoretically that there is little difference between smaller and larger laser pulse width than time bin cases on measurements of both detection and false-alarm probability.

1. Introduction

There are various methods for optical distance measurements (interferometry, time-of-flight and triangulation methods) [1]. For higher ranging capability, the laser radar system that measures TOF of laser pulse with the Geiger mode APD as detector has been developed by a few research groups due to its extremely high sensitivity [2–4]. The APD which is reverse biased above the breakdown voltage to operate in photon-counting mode, is the Geiger mode in which the primary electron generated by the absorption of a single photon initiates an avalanche process. The avalanche process constitutes the electrical current surge which has a sharp leading edge, allowing the high resolution timing [5].

However there are some negative aspects on Geiger mode APD as detector of the laser radar. First, the dark counts generated by thermal noises in the depletion region can cause the false-alarms. Second, the Geiger mode APD has the dead time after the detecting the photon in which the detector will not respond to any photons. In other words, the Geiger mode APD takes the time to reset after detecting and making the current surge which is the signal announcing the arrival of photons. The dead time typically varies from 10 ns to 1 μs depending on which material is used and how quenching circuit is designed. Assuming tens of kilohertz laser pulse repetition rate, multiple signals can be detected during the gate time in the case of short dead time (multi hit case), while only one signal can be detected during the gate time in the case of relatively long dead time (single hit case). If the Time to Digital Converter (TDC) can record only the first signal during the gate, then this laser radar belongs to single hit case even though the Geiger mode APD is capable of detecting multiple signals. In this paper only the single hit case is dealt in order to focus on the proving the theory that the generation of primary electrons in the Geiger mode APD follows the Poisson statistics.

In direct-detection laser radar, the number of receiver integrated primary electrons from a diffuse target follows a negative-binomial distribution [6,7]. When the number of photons received is much less than the speckle diversity, the Poisson distribution is a good approximation to the negative-binomial distribution [8].

In the next section, the direct-detection laser radar system with longer laser pulse than time bin is modeled with the assumption that the creations of primary electrons in the Geiger mode APD is Poisson distributed. It is shown that the calculated detection and false-alarm probabilities mostly do not depend on laser pulse width a few times larger than time bin, so that the experimental setup in Section 3 is regarded as appropriate. Section 3 explains the experimental setup for Poisson statistics on TOF measurements with Geiger mode APD and shows the results. The last section is summary.

2. Modeling

The mean number of primary electrons generated by single laser pulse scattered from the target is calculated analytically with laser range equation. And the mean number of primary electrons generated by noises including background lights and dark counts in the Geiger mode APD is assumed to be constant. Then, the Poisson statistics is used to derive the detection and false-alarm

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probabilities for Geiger mode APD considering laser pulse width larger than the time bin.

2.1. Laser range equation

The mean number of photons impinge on detector is the sum of laser signal photons and background photons composed of the sunlight both reflected from target and scattered by atmosphere in the field of view of laser radar system, and blackbody radiation from the target. The laser range equation model is illustrated in Fig. 1.

When transmitting a laser pulse of energy, E_T, the mean number of photons returning from a flat, diffusely reflecting extended target (Lambertian target) at distance R with the reflectivity ρ, $S_{P\text{-total}}$ is

$$S_{P\text{-total}} = E_T \frac{\lambda}{hc}\left(\frac{FOV}{\theta_T}\right)^2 \frac{\rho}{\pi} \cos\theta_{\text{target}} \frac{A_R}{R^2} \eta_T \eta_R \eta_A^2, \quad (1)$$

where

$$E_T = \int_{\tau_{\text{width}}} P_T(t)\, dt, \quad (2)$$

P_T is the transmitting optical power; τ_{width} is the width of laser pulse; h is Plank's constant; λ is the wavelength of laser pulse; c is the speed of light; FOV is the field of view of receiver; θ_T is the divergence angle of laser beam; θ_{target} is the angle of incidence of the laser beam relative to the surface normal; A_R is the area of the aperture of receiver; η_T is the transmission of transmitter; η_R is the transmission of receiver optics; and η_A is the one-way transmission of atmosphere between the target and receiver [9]. Let τ_{bin} be the time bin and S_{PE} be the mean number of generated primary electrons by laser pulse scattered by the target during τ_{bin}. Then S_{PE} is

$$S_{\text{PE}} = \int_{\tau_{\text{bin}}} P_T(t)\, dt \eta_Q \frac{\lambda}{hc}\left(\frac{FOV}{\theta_T}\right)^2 \frac{\rho}{\pi} \cos\theta_{\text{target}} \frac{A_R}{R^2} \eta_T \eta_R \eta_A^2, \quad (3)$$

where η_Q is the quantum efficiency of the Geiger mode APD.

In this paper, background photons rate is assumed to be constant so that during the gate time the mean number of generated primary electrons by background noise, N_{PE} is

$$N_{\text{PE}} = \tau_{\text{gate}}(\eta_Q N_{\text{BG}} + f_{\text{dark}}), \quad (4)$$

where N_{BG} is the solar background photons rate assumed to be constant and f_{dark} is the dark counts rate of the Geiger mode APD.

2.2. Poisson statistics

For a Poisson process, the probability that m events occur during times t_1 and t_2 is

$$P(m; t_1, t_2) = \frac{1}{m!}[M(t_1, t_2)]^m \exp[-M(t_1, t_2)], \quad (5)$$

where

$$M(t_1, t_2) = \int_{t_1}^{t_2} r(t)\, dt, \quad (6)$$

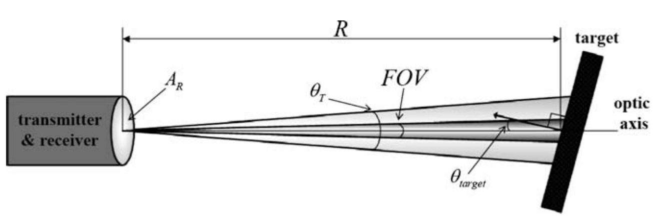

Fig. 1. The laser range equation model.

and $r(t)$ is the rate function of the process [8]. In the case of laser radar using Geiger mode APD, the creation of primary electron is the event and the mean rate of generation of primary electrons created by in coming photons between times t_1 and t_2. From Eq. (5), the probability that no primary electrons are created between times t_1 and t_2 is $\exp[-M(t_1, t_2)]$ and the probability that one or more are created is $1 - \exp[-M(t_1, t_2)]$.

2.3. Detection and false-alarm probabilities

In the case of smaller laser pulse width than time bin, the single pulse detection probability at distance R, $P_{\text{detect-small}}$ is

$$P_{\text{detect-small}} = \exp(-fn)[1 - \exp(-S_{\text{PE-total}} - n)], \quad (7)$$

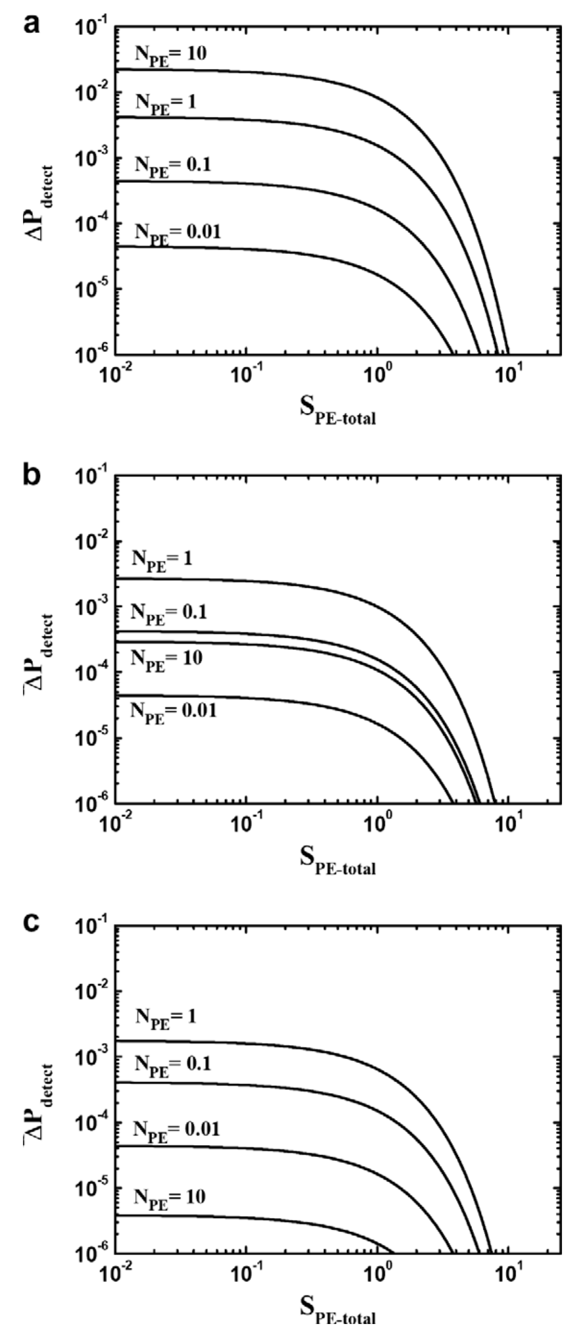

Fig. 2. The difference between single pulse detection probabilities of shorter and longer laser pulse cases versus the total number of primary electrons generated by laser signal. The curves are differentiated by, and labeled with, the N_{PE} in units of primary electrons per gate interval. (a) R = 10 m, (b) R = 75 m and (c) R = 145 m.

where

$$n = N_{PE}\frac{\tau_{bin}}{\tau_{gate}}, \tag{8}$$

$$S_{PE\text{-total}} = \eta_Q S_{P\text{-total}}, \tag{9}$$

$$f = \frac{2R}{c\tau_{bin}}, \tag{10}$$

and the single pulse false-alarm probability, $P_{false\text{-small}}$ is

$$P_{false\text{-small}} = 1 - P_{detect\text{-small}} - \exp(-S_{PE\text{-total}} - N_{PE}), \tag{11}$$

where τ_{gate} is the gate time [10].

In the case of the laser pulse having larger pulse width than time bin but the same energy, let the laser pulse width be

$$\tau_{width} = l\tau_{bin} \tag{12}$$

where l is integer and the instantaneous optical power of laser pulse is constant during τ_{width} for the simplicity. Then the single pulse detection probability for larger pulse, $P_{detect\text{-large}}$ is

$$P_{detect\text{-large}} = \sum_{i=f+1}^{f+l} P_i = \exp(-fn)[1 - \exp(-S_{PE\text{-total}} - ln)], \tag{13}$$

where

$$P_i = \exp(-fn)\exp[-(i-(f+1))(S_{PE}+n)][1-\exp(-S_{PE}-n)], \tag{14}$$

and the single pulse false-alarm probability for larger pulse, $P_{false\text{-large}}$ is

$$P_{false\text{-large}} = 1 - P_{detect\text{-large}} - \exp(-S_{PE\text{-total}} - N_{PE}). \tag{15}$$

In order to compare both cases, let

$$\Delta P_{detect} = P_{detect\text{-large}} - P_{detect\text{-small}} = -\Delta P_{false}. \tag{16}$$

Fig. 2 shows the ΔP_{detect} in the case that l = 10, τ_{bin} = 0.5 ns and τ_{gate} = 1 μs. Using the larger pulse width, the detection probability is improved by background noise less than 1% except when R = 10 m and N_{PE} = 10. The results show that there is little difference on detection and false-alarm probabilities between the larger and the smaller laser pulse width cases.

3. Experiments

Fig. 3 shows the diagram of the optical system for measuring the TOFs of repetitive laser pulses of laser radar system using Geiger mode APD as detector in the laboratory.

The passively Q-switching microchip laser of which wavelength is centered on 532 nm and pulse width is 5 ns, is used as light source and optical bandpass filter blocks both the 1064 nm and the 808 nm [11]. Because of the single polarization of the laser, HWPs are located before the PBS in order to control both the transmission and the reflection of the laser pulse at the PBSs. The start signals are not able to be generated by laser itself because the laser is passively Q-switched. Therefore the S-polarization of laser pulse is reflected to the fast PD in the first PBS to generate the electrical start signals and this start signals initiate the TDC. After the first PBS, the energy of laser pulse is controlled by both the variable ND filter and the second HWP. The transmitted laser pulse is headed to the target and scattered. The diffuse reflectance standard of Labsphere is used as the Lambertian target. The scattered light in *FOV* is collected into Geiger mode APD through the QWP, the second PBS, the OBF and focusing lens in turn. The Geiger mode APD, Id Quantique Id100-20-ULN of which maximum count rate and dark count rate are 20 MHz and less than 1 Hz respectively, collects both the laser pulse scattered from the target and the noises through the optical setup in Fig. 3 [12]. Agilent 53131A of which timing resolution is 500 ps, is used as TDC. The Hg light, the Halogen lamps and the fluorescent lamps in laboratory are used in order to generate the continuous noise artificially.

3.1. Continuous noise source generated artificially

With continuous noise source, the probability which the generation of primary electron in Geiger mode APD occurs in the bin right after t times, $P_{detect\text{-cw}}$ is

$$P_{detect\text{-cw}}(t) = \exp\left(-\frac{N_{PE}}{\tau_{gate}}t\right)[1-\exp(-n)], \tag{17}$$

Hg light is used as continuous light source to prove that the Geiger mode APD of direct-detection laser radar system is governed by Poisson process. Blocking the laser pulse before the second PBS and illuminating the target with Hg light, the Geiger mode APD collects only Hg light scattered from the target through the optical setup so that the detection timings of continuous light Hg during every gate intervals are measured. With 3 kHz trigger signal, 90,000 time intervals are measured and divided into 100 ns time bins.

Fig. 4 shows the results and the fitting function is

$$P_{detect\text{-cw}}(t) = A\exp(-t/B) + C, \tag{18}$$

and in the case above A = 0.18959, $B = 4.1882 \times 10^{-7}$ and C = 0.00173 with standard error 0.00308, 1.00408×10^{-8} and

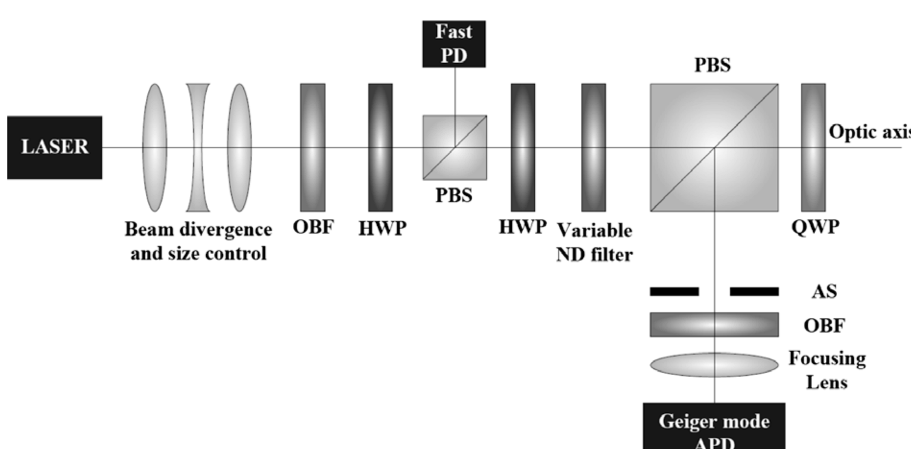

Fig. 3. Optical setup for measurements of the single pulse detection and false-alarm probabilities. OBF, optical bandpass filter; HWP, half-wave plate; PBS, polarization beam splitter; ND filter, neutral density filter; QWP, quarter-wave plate; AS, aperture stop; PD, photodiode.

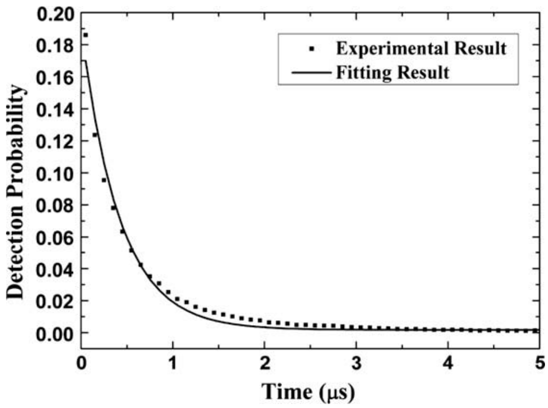

Fig. 4. Detection probability versus time with Hg light, continuous noise source generated artificially. The dots are experimental results and the straight line is their fitting curve.

3.36952×10^{-4} respectively. In the case of even smaller time bins, the detection probability decaying exponentially is acquired with the same B value in Eq. (17) and lower detection probability values. It is clear that the Geiger mode APD, used in this experiment, is governed by Poisson process. In the next subsection, this method is used to measure the number of primary electrons generated by both background lights and dark counts during the gate time. In the continuous noise case, the B value in Eq. (17) is the inverse of rate function which means the generation rate of primary electrons by noises.

3.2. Measurements of detection and false-alarm probabilities

In this subsection, the theoretical models (Eqs. (13) and (15)) are proved systematically with the experimental setup in Fig. 3. At first, blocking the laser pulse before the second PBS and illuminating the target with continuous noise source, the generation rate of primary electrons by the noise is determined. Next, controlling the energy of laser pulse by rotating the second HWP and changing the ND filter, the TOF data of 1000 laser pulses are acquired by TDC per each case of different energy. Then, the detection probability is determined by dividing the number of TOF data in the target bins by the total number of laser pulses and the false-alarm probability is determined by dividing the number of the rest of TOF data in the gate by the total number of laser pulses. The experimental results are represented in Fig. 5. The number of primary electrons generated by scattered laser pulse from the target is derived by Eq. (3) where $h = 6.63 \times 10^{-34}$ J s, $\lambda = 532$ nm, $c = 3 \times 10^{8}$ m/s, $c = 3 \times 10^{8}$ m/s, $\theta_{target} = 0$ rad, $\tau_{bin} = 0.5$ ns, $FOV = 0.4$ mrad, $\theta_T = 6$ mrad, $\eta_A = 10^{-a(\lambda)R/10{,}000} = 0.996644$, $a\,(\lambda = 532\,nm) = 0.73$ dB/km, $\eta_R = 0.2576$ and $\eta_Q = 0.35$ [13]. Because the energy of transmitted laser pulse is calculated by measuring both the optical power after the QWP and the repetition rate of the laser, $\eta_T = 1$.

The results show that the derived Eqs. (13) and (15) with Poisson statistics are well matched with the experimental data.

4. Summary

It is shown that in the case of direct-detection laser radar using Geiger mode APD, with the same pulse energy, the single pulse detection and false-alarm probabilities seldom depend on laser

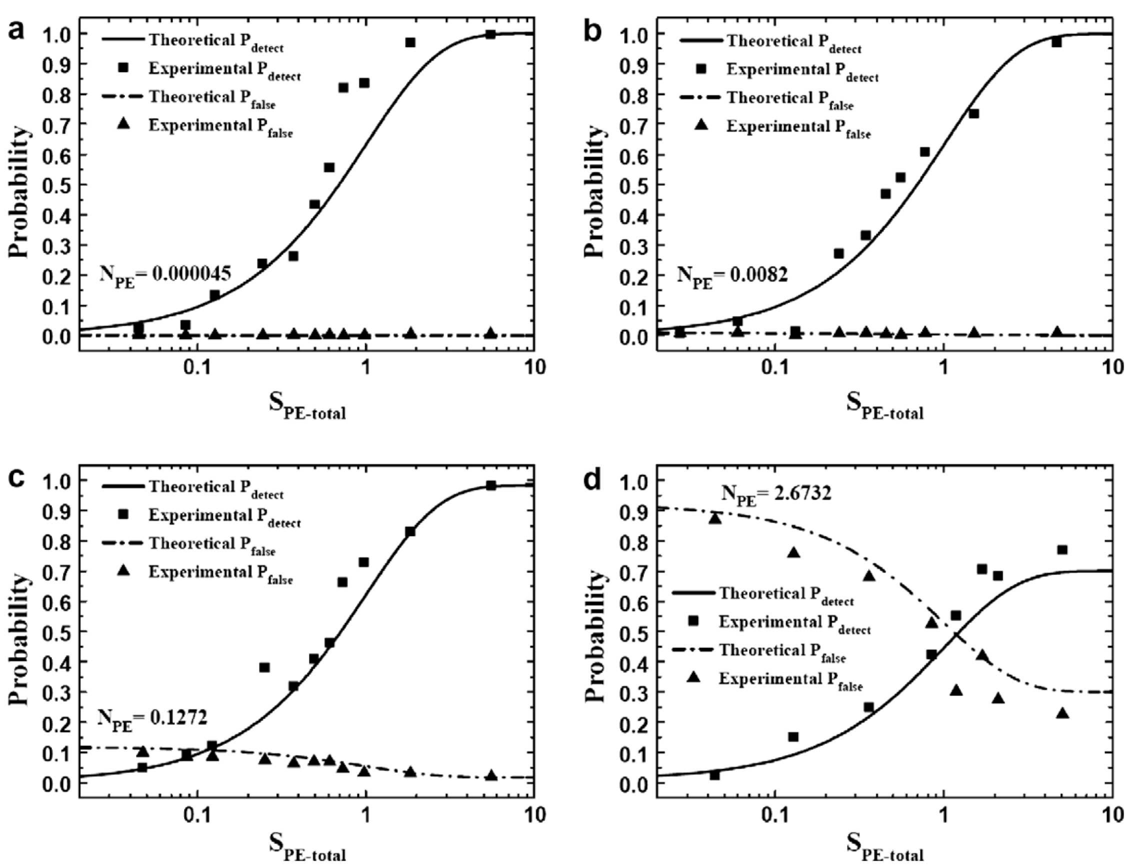

Fig. 5. Single-pulse detection (solid curve) and false-alarm (dashed curve) probabilities acquired by theory (dots) and experiment (curve) versus the number of primary electrons generated by laser pulse in the case of (a) $N_{PE} = 0.000045$, (b) $N_{PE} = 0.0082$, (c) $N_{PE} = 0.1272$ and (d) $N_{PE} = 2.6732$.

pulse width in Section 2.3. Then, with the larger laser pulse width than time bin, the experiment for the proof of Poisson statistics on direct-detection laser radar using Geiger mode APD is successfully accomplished by measuring the single pulse detection and false-alarm probabilities with continuous noise source generated artificially. This result is expected to enhance the Poisson model on direct-detection laser radar using Geiger mode APD and be used to design or optimize the laser radar system.

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References

[1] Markus-Christian Amann, Thierry Bosch, Marc Lescure, Risto Myllylä, Marc Rioux, Laser ranging: a critical review of usual techniques for distance measurement, Opt. Eng. 40 (1) (2001) 10–19.
[2] Marius A. Albota, Richard M. Heinrichs, David G. Kocher, Daniel G. Fouche, Brian E. Player, Michael E. O'Brien, Brian F. Aull, John J. Zayhowski, James Mooney, Berton C. Willard, Robert R. Carlson, Three-dimensional imaging laser radar with a photon-counting avalanche photodiode array and microchip laser, Appl. Opt. 41 (36) (2002).
[3] Vincent Delaye, Pierre Labeye, High-resolution eye safe time of flight laser range finding, Proc. SPIE 4035 (2000) 216–225.
[4] Johnson Steben, Philip Gatt, Terry Nichols, Photon counting imaging: the Digital APD, Proc. SPIE 6068 (2006).
[5] Brian F. Aull, Andrew H. Loomis, Douglas J. Young, Richard M. Heinrichs, Bradley J. Felton, Peter J. Daniels, Deborah J. Landers, Geiger-mode avalanche photodiodes for three-dimensional imaging, Lincoln Lab. J. 13 (2002) 335–350.
[6] P. Gatt, S.W. Henderson, Laser radar detection statistics: a comparison of coherent and direct detection intensity receivers, Proc. SPIE 4377 (2001) 251–262.
[7] J.W. Goodman, Some effects of target-induced scintillation on optical radar performance, Proc. IEEE 53 (1965) 1688–1700.
[8] J.W. Goodman, Statistical Optics, Wiley, New York, 1985.
[9] Albert V. Jelalian, Laser Radar Systems, Artech House Boston, London, 1992.
[10] Daniel G. Fouche, Detection and false-alarm probabilities for laser radars that use Geiger-mode detectors, Appl. Opt. 42 (27) (2003).
[11] J.J. Zayhowski, Passively Q-switched microchip lasers and applications, Rev. Laser Eng. 26 (1998) 841–846.
[12] Id Quantique, id100, <http://www.idquantique.com/products/files/id100-specs.pdf>.
[13] RCA, Electro-Optics Handbook, Radio Cooperation of America, Harrison, NJ, Commercial Engineering.